

Experiment No: 5

Title : Implementation of different types of Joins

- Inner Join
- Outer Join
- Natural Join..etc

Objective :

To implement different types of joins

Theory :

The SQL **Joins** clause is used to combine records from two or more tables in a database. A JOIN is a means for combining fields from two tables by using values common to each. The join is actually performed by the 'where' clause which combines specified rows of tables.

Syntax:

```
SELECT column 1, column 2, column 3...  
FROM table_name1, table_name2  
WHERE table_name1.column name = table_name2.columnname;
```

Types of Joins :

1. Simple Join
2. Self Join
3. Outer Join

Simple Join:

It is the most common type of join. It retrieves the rows from 2 tables having a common column and is further classified into

Equi-join :

A join, which is based on equalities, is called equi-join.

Example:

```
Select * from item, cust where item.id=cust.id;
```

In the above statement, item-id = cust-id performs the join statement. It retrieves rows from both the tables provided they both have the same id as specified by the where clause. Since the where clause uses the comparison operator (=) to perform a join, it is said to be

equijoin. It combines the matched rows of tables. It can be used as follows:

- ✓ To insert records in the target table.
- ✓ To create tables and insert records in this table.
- ✓ To update records in the target table.
- ✓ To create views.

Non Equi-join:

It specifies the relationship between columns belonging to different tables by making use of relational operators other than '='.

Example:

```
Select * from item, cust where item.id<cust.id;
```

Table Aliases

Table aliases are used to make multiple table queries shorter and more readable. We give an alias name to the table in the 'from' clause and use it instead of the name throughout the query.

Self join:

Joining of a table to itself is known as self-join. It joins one row in a table to another. It can compare each row of the table to itself and also with other rows of the same table.

Example:

```
select * from emp x ,emp y where x.salary >= (select avg(salary) from x.emp  
where x. deptno =y.deptno);
```

Outer Join:

It extends the result of a simple join. An outer join returns all the rows returned by simple join as well as those rows from one table that do not match any row from the table. The symbol(+) represents outer join.

- Left outer join
- Right outer join
- Full outer join

LAB PRACTICE ASSIGNMENT:

Consider the following schema:

Sailors (sid, sname, rating, age)

Boats (bid, bname, color)

Reserves (sid, bid, day(date))

1. Find all information of sailors who have reserved boat number 101.
2. Find the name of boat reserved by Bob.
3. Find the names of sailors who have reserved a red boat, and list in the order of age.
4. Find the names of sailors who have reserved at least one boat.
5. Find the ids and names of sailors who have reserved two different boats on the same day.
6. Find the ids of sailors who have reserved a red boat or a green boat.
7. Find the name and the age of the youngest sailor.
8. Count the number of different sailor names.
9. Find the average age of sailors for each rating level.
10. Find the average age of sailors for each rating level that has at least two sailors.